# SAWSTAR

## User Manual

Simple Synth - Big Sound
The Sounds of Trance




## About this RC guide

This is the unpublished SAWSTAR 1.2.1-rc1 test-candidate guide, not a final release or native acceptance report. It covers 30 factory presets. Screenshots remain from 1.0.3, capture build 8040390; they illustrate the established layout. Read chapter 15 before replacing a plugin. Edition: development.

## Reading the interface

Names such as MAIN, ADVANCED, PRESETS, Save As and NOISE COLOR are the labels to look for in the plugin. Start with MAIN for sound design; ADVANCED contains performance and deeper controls; PRESETS manages the library and explains saved sounds.

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# 02 / Installation and first sound

## RC packages

The Premium integration RC is prepared as macOS Universal and Windows x64 VST3 bundles. Windows ARM64 Premium RC is not qualified. The build does not install automatically. Use only a package with matching source, hashes and validator evidence; a successful build does not imply host acceptance.

## RC installation

Keep the previous plugin and project backups. Close the DAW before replacing the complete SAWSTAR.vst3 bundle in the standard VST3 directory. There is no final signed installer for this RC. Do not install it beside Classic under the same identity. Start with a new test project or a copy, rescan if needed, and verify About before opening important sessions.

## Unsigned packages

macOS packages have no Apple Developer ID signature or notarization; Windows installers are not Authenticode signed. Security warnings may appear. Follow the package INSTALLATION guide and approve only the trusted SAWSTAR download. Do not disable system-wide protections.

## First sound in REAPER

Insert SAWSTAR as an instrument on a track. Click a key on its onscreen keyboard. For an external keyboard, choose the track MIDI input, arm the track and enable input monitoring as appropriate. Check that the track and master are unmuted and routed to your audio device. Select SuperSaw One in the top preset selector and play a few notes.

## Updating safely

Back up user presets and important projects before updating the plugin. Check the About build information after restarting the DAW: an open host may retain the old plugin. Factory presets are embedded and need no separate bank installation.

# 03 / Navigation and controls

## Three views, one instrument

MAIN, ADVANCED and PRESETS share the top preset selector and the bottom keyboard, wheels and status strip. Switching views does not load a different sound. The preset selector and its arrows change the loaded preset. The gear menu provides interface scaling, the downloads link and About; OK closes About.

## Editing a sound

Drag knobs and faders to change values. Use the value-entry field where available for an exact number; confirm the entry with Enter. Dropdown controls choose an option. The blue active state distinguishes enabled switches from their darker inactive state. Changing a control affects the sound immediately; save the sound if you want to keep it.

## Keyboard and wheels

The compact keyboard covers MIDI notes 36-96; the host can send the full MIDI note range. There is no octave-switch control for the onscreen keyboard. PITCH springs back to center when released. MOD retains its position. Double-click either wheel to reset it. The GUI keyboard and wheels use MIDI channel 1; external MIDI controller processing is channel-aware.

## What is saved

Sound parameters belong to presets and DAW state. Wheel positions are temporary MIDI controller values: record wheel movements as MIDI in your DAW. The footer reports sample rate, active voices and CPU information. These are status indicators, not fixed performance promises.



# 04 / Sound sources and output

## OSC 1 and OSC 2

Choose Saw, Square, Triangle or Sine in the waveform selector. OCT transposes the oscillator by octaves. DETUNE spreads the side oscillators. UNISON blends six detuned side oscillators with the central oscillator; it is not a polyphony selector. At zero, the center dominates. WIDTH spreads the unison across stereo. Start with a saw, moderate UNISON and a small DETUNE value, then adjust by ear.

## SUB and MIXER

SUB offers Sine, Triangle and Square with a lower octave range. Its volume is controlled by the SUB mixer fader. The four mixer faders balance OSC 1, OSC 2, SUB and NOISE. A waveform can be configured but remain inaudible if its mixer level is zero.

## Noise

Choose White, Pink or Dark noise below the mixer. White is bright and broad, Pink emphasizes lower frequencies relative to White, and Dark provides a darker character. NOISE COLOR further shapes the tone: negative values darken it and positive values brighten it by reducing low-frequency content. The NOISE fader controls how much noise enters the sound.

## OUTPUT

VOLUME sets the master output level; BOOST adds gain. Watch the output meter and listen while balancing loud presets. WIDE and its AMOUNT expand existing stereo separation after the effects; they do not create stereo from a mono signal. Oscillator WIDTH works earlier in the signal path. Check mono compatibility when using wide sounds. Internal peak protection does not guarantee that later DAW effects or summed tracks will never clip.

# 05 / Filter, envelopes and scope

## FILTER

Choose Low Pass 12, Low Pass 24, High Pass 12 or Band Pass 12. CUTOFF sets the filter frequency, RES emphasizes the cutoff region and DRIVE adds filter character. FILTER MIX blends the filtering into the sound: at zero, a cutoff change may be inaudible. KEY TRACK makes cutoff follow played pitch. The cutoff display is an illustration of the filter setting.

## Two envelopes

AMP ENV controls the loudness of each note. A is attack time, D is decay time, S is the held sustain level, and R is release time after letting go. FILTER ENV has its own ADSR and AMOUNT: positive amount opens the filter, negative amount moves it in the opposite direction. A short decay and low sustain are useful for plucks; longer attack and release suit pads.

## Realtime waveform

The green scope below AMP ENV shows the summed voice signal before effects, master width and output gain. Its horizontal axis is time; the center line is zero amplitude, not 0 Hz. The dark grid and triggered display help reveal waveform structure. The display scales amplitude automatically, so it is not a calibrated loudness meter.

## Interpreting the trace

A single periodic note is easier to read than a detuned chord or noise. A complex moving trace is normal for those sounds. Stereo is combined for this display, so cancellation can reduce the trace. Changing OUTPUT or adding delay will not change this pre-effects view. The scope is visual only and does not process the audible signal.

# 06 / Effects

## Signal order

The voice signal reaches Chorus, then Delay, then Reverb, followed by master stereo width and output processing. The MAIN FX section provides the quick controls; open the corresponding effect on ADVANCED for detailed settings. Enable an effect and raise its MIX to hear it.

## Chorus

Chorus adds movement and thickness using a modulated delayed signal. MIX balances wet and dry sound. RATE controls movement speed and DEPTH its extent. For a steady lead, use a modest mix and slow rate; excessive depth can make pitch movement more obvious.

## Delay

MIX sets the echo level. TIME selects milliseconds in free mode; tempo sync uses a musical DIVISION and the host tempo. FEEDBACK sets the number and persistence of repeats, while TONE darkens or brightens them. Stereo and Ping Pong modes give different spatial patterns. Lower feedback and mix if echoes obscure the next phrase.

## Reverb

MIX adds the reverberant signal. SIZE changes the space character, DECAY controls tail duration and DAMPING shapes the high-frequency decay. Use shorter, quieter reverb for rhythmic plucks and more space for pads. Long release, delay feedback and reverb tails can all continue after a key is released; this is not automatically a stuck-note fault.

# 07 / Performance and arpeggiator

## Voice modes and glide

Poly plays chords with up to 16 synth voices. Mono and Legato are single-note performance modes; Legato suits connected phrases. GLIDE controls the transition between pitches. Its mode can apply it always or only when notes overlap. PITCH BEND range is adjustable up to 24 semitones; MOD DEPTH controls the built-in modulation-wheel route to cutoff.

## Build an arpeggio

Enable ARP and hold a chord. ORDER chooses Up, Down, Up/Down, Random or Played. Played follows key order. RATE selects the musical step length; OCTAVES extends the pattern across 1-4 octaves. GATE controls how much of a step the note lasts. SWING alternates longer and shorter steps.

## HOLD

HOLD latches the chord after your fingers leave the keys. After all physical keys are released, playing a new chord replaces the latched phrase. Disable HOLD to stop relying on the latch; sustain-pedal behavior can still affect held notes. Disable ARP to return to ordinary playing.

## Tempo behavior

The first held key starts the phrase. The arp follows host BPM but is not an absolute song-grid sequencer. It can play while the host is stopped; stopping a running host clears the phrase, and a new key starts it again. Generated notes drive SAWSTAR internally; the arp is not presented as a MIDI-output or programmable step-sequencer feature.

# 08 / LFO and modulation

## Two independent LFOs

LFO 1 / LFO 2 selects which LFO you edit; both can operate simultaneously. Choose Sine, Triangle, Ramp or Square. RATE uses free frequency or tempo-synced division. Retrigger restarts the phase on the first key; free phase keeps it moving. The direct destination can be cutoff, pitch, amplitude or pan. AMOUNT sets the direct route depth.

## Four modulation rows

Each row selects a source, a destination and a signed amount. Sources are Off, LFO 1, LFO 2, Mod Wheel, Velocity and Aftertouch. Destinations include Filter Cutoff, Pitch, Amp Level, Pan and Noise Color. Negative amount reverses the direction. Aftertouch means MIDI channel pressure; polyphonic key pressure and MPE are not supplied here.

## Avoid accidental double modulation

An LFO matrix source is taken before its direct AMOUNT. Therefore, setting direct AMOUNT to zero does not silence an active matrix route. Multiple routes add together. MOD DEPTH provides a separate built-in wheel-to-cutoff route, even when all matrix rows are Off. Wheel down adds zero cutoff offset: it is the preset baseline, not an incorrect or incomplete sound. Raising the wheel opens the filter according to MOD DEPTH. A Mod Wheel matrix row adds its own effect on top. Set MOD DEPTH to zero if the matrix should provide the only wheel assignment.

## Peak meter

The stereo sample-peak meter has a one-second peak hold. CLIP lights indicate output samples reaching 0 dBFS. Click to clear, or wait five seconds after the last clip. Output protection normally prevents clipping. This is not RMS or true-peak metering.

## A subtle pad example

Select a pad. Set LFO 2 to a slow sine, with direct AMOUNT zero. In one matrix row choose LFO 2, Pan and a small positive amount. Listen in stereo, then reduce the row amount to compare. Save As preserves the result as a new user preset.

# 09 / Preset workflow

## Browse and understand

Click a preset row to load it. Category and search filters narrow the list without changing the sound by themselves. Search matches names, categories and tags. The heart marks a favorite. HOW IT WORKS shows the saved preset settings, not your unsaved edits; it updates when the edited sound is saved.

## Save and Save As

Save overwrites the active changed user preset after confirmation. Factory presets and templates are read-only: use Save As to create a user copy. Save As opens a naming/file dialog and creates a new .sawstar file; use a different name when a file already exists. An externally saved file is also copied into the managed library when possible. A collision can prevent that library copy.

## Import, Copy and Paste

Import accepts multiple .sawstar files at once, so a collection need not be imported one file at a time. It reports skipped or failed items and does not automatically play a preset. Copy/Paste transfers sound settings within the editor and creates no file until saved. This clipboard is temporary and is cleared when the editor closes.

## Rename, Delete and Initialize

Rename changes a selected user preset name. Delete asks for confirmation and retains a .deleted backup; it does not stop the currently playing sound. Initialize asks before resetting the sound to Init. Save changes you want to keep before initializing or loading another sound.

## Library and recovery

User files: macOS ~/Library/Application Support/SAWSTAR/Presets ; Windows %APPDATA%\SAWSTAR\Presets . Save backups are kept in .sawstar-backups below that folder. Restore a backup by copying it into the library under an unused .sawstar filename. DAW projects store sound parameters independently, but may not restore the association with a user file for Save; reload that user preset to re-establish it.

# 10 / Factory library: 30 presets

## Seven Templates

Init • Pad-Init • Lead-Init • Pluck-Init • Bass-Init • Keys-Init • Arp-Init. These are deliberate starting points rather than finished demonstrations. Effects and movement are kept off. In Arp-Init, enable ARP yourself to begin developing a pattern.

## Leads

Wide Saw Lead • SuperSaw One • Silk Lead • Solid Saw Lead. Explore stereo unison, fuller saw layers and connected lead playing. Compare with Lead-Init to understand which changes create the finished character.

## Pads

Soft Pad • Soft Pad One • Warm Horizon • Velvet Pad. Use held chords to hear envelope development and tails. Keep space for other instruments when adding width and reverb.

## Plucks, basses and keys

Plucks: Saw Pluck • Glass Pluck • Trance Pluck. Basses: Deep Bass • Round Sub. Keys: Bright Keys • Air Keys. Try short and long notes, plus different velocities, to understand the envelope response.

## Arpeggiated sounds

Simple Steps • Octave Steps. Hold a chord and explore rate and gate before changing the oscillators. Empty categories in the browser do not indicate a missing installation; this edition contains 30 presets, including the six performance sounds below. User presets are additional and are not part of the embedded factory count.

## Six performance sounds

Wheelfire (use modwheel): LP24 legato lead. Velvet Orbit (use modwheel): moving BP12 pad. Copper Pulse (use modwheel): synced LP24 bass. Polar Keys (use modwheel): BP12 keys. Neon Current (hold chord): LP12 up/down arp. Glass Circuit (hold chord): HP12 descending arp. Wheel opens the filter; HOLD stays off. Robert approved these six sounds using the separate preset test pack; exact-build host acceptance remains open.

# 11 / Three sound design exercises

## A broad lead

Load Lead-Init. Choose Saw for OSC 1, raise UNISON gradually and use moderate DETUNE. Bring in OSC 2 at a lower mixer level to add body. Adjust WIDTH while checking mono. Use a fast amp attack and a release that does not overlap phrases too much. Add a restrained delay, then balance OUTPUT against another instrument. Save As with your own name.

## A soft pad

Load Pad-Init. Blend two oscillators and lengthen amp attack and release. Lower cutoff with FILTER MIX active, and use a gentle filter-envelope amount. Add a small chorus and reverb. If needed, introduce a very slow, low-depth pan or cutoff LFO. Avoid adding modulation only because it is available: a stable pad is a valid design.

## A clear pluck

Load Pluck-Init. Use a short filter-envelope decay, low sustain and a positive filter amount. Match the amp envelope to the desired note length. Add a quiet tempo-synced delay and little reverb so the attack remains clear. Play repeated notes at the song tempo before saving.

## Compare fairly

A louder setting can seem more impressive without being better. Match output levels when comparing variants. Change one section at a time, save useful variations, and use HOW IT WORKS to revisit the saved signal-path choices.

# 12 / Troubleshooting and support

## No sound

Check host audio output, track routing and monitoring, mixer source levels and OUTPUT. Disable ARP/HOLD for a simple key test. A filter with very low cutoff and full mix can strongly attenuate the sound. Try Init or a known factory preset.

## A control seems ineffective

Check the relevant enable switch and MIX. DETUNE and WIDTH have little audible effect when unison side voices are not mixed in. WIDE needs stereo content. The scope is before effects. HOW IT WORKS follows saved settings. An LFO may act through a matrix row even with its direct amount at zero.

## A sound continues

Check amp release, sustain pedal, ARP HOLD, delay feedback and reverb decay. If sound continues unexpectedly after these are cleared, stop playback and reload the instrument in a test project. Report the exact preset, MIDI sequence, host and build information rather than overwriting an important project.

## Preset or display issue

Clear search/category filters if a preset seems missing. Reload a user preset before Save if its association was lost after project recall. Try another GUI scale if the window does not fit. Reopen the editor after an update, and verify the About build. Imported malformed or conflicting files can be skipped; read the import result.

## Project and credits

SAWSTAR: Simple Synth - Big Sound. The Sounds of Trance. Created by Gyuricza Róbert / RobCZart82. Project, source and issue reporting: https://github.com/RobCZart82/SAWSTAR . The project license and third-party notices are supplied in the repository. See the release notes for download packages and supported configurations.

# 13 / Performance details

## Mono and Legato

These modes select one note using last-note priority; physically held keys take priority over sustain-latched keys. Releasing the selected key may return to an older held key. Changing voice mode starts a new key phrase; release and play the keys again. A short pitch return can be audible and is not proof of a corrupt preset.

## LFO retrigger and event order

Retrigger restarts the global LFO when a first key arrives. Same-time MIDI events retain their incoming order. Note Off then Note On can restart the LFO, while overlapping Note On then Note Off can preserve its phase. Free phase avoids this retrigger distinction. This behavior is retained for compatibility.

## MIDI reset messages

CC121 resets pitch bend, modulation wheel and channel pressure and releases sustain on its channel; controller smoothing can take time to settle. CC120 stops sound on its channel but preserves wheel and bend values. Sending CC120 on all 16 channels also clears shared effect tails. A new note can then start normally. CC123 releases notes; sustain and release tails can still matter.

## Live edits and tails

Changing envelope times during playback affects the active sound. Preset or effect changes can also change an existing tail; compare sounds after the previous tail has finished. Save valuable sounds before changing presets.

## Meter and update checks

CLIP observes protected output samples, not overload inside an earlier processing stage. It is not an inter-sample true-peak indicator. After updating, check the About version in the actual host. The screenshots show version 1.0.3, capture build 8040390. Installation and removal details: github.com/RobCZart82/SAWSTAR/blob/main/docs/INSTALLATION.md

# 14 / Earlier compatibility fixes

## Editor and MIDI lifecycle

Closing the editor releases its mouse-held keyboard note and completes an active PITCH-wheel gesture. MOD remains latching; closing the window is not an all-channel panic. Reset and editor-overflow recovery distinguish editor notes from host-owned notes. GUI wheels deliver the latest state at the next real block; intermediate gestures can be combined, while host MIDI retains its sample offsets.

## Bypass and release tails

Bypass keeps the MIDI/engine timeline running but mutes host output. Unbypass resumes the current sound instead of replaying stale input. The plugin reports a conservative 380-second finite tail covering maximum serial envelope and effect history. This is a rendering bound, not a forced note timeout. A host may append long silence when using this bound; choose its render-tail or trimming settings deliberately.

## Preset file validation

A .sawstar file must have the versioned SAWSTAR header, non-empty known settings and the exact declared size. Headerless data, empty/unknown-only files and host trailers are rejected. Old versioned partial presets still load; DAW projects keep their legacy migration. Enum/integer values round to the same values the host applies, so loading a fractional enum does not falsely mark the preset modified.

## Update and verify

Close the DAW before replacing the plugin and back up projects and user presets. Reopen a project copy and check About for 1.2.1-rc1 and the actual candidate/commit. RC builds are test candidates, not the final release. The established layout, parameter IDs and state format remain compatible. Screenshots in this guide are explicitly from 1.0.3; they illustrate the same layout.

# 15 / Premium integration RC

## Engine and identity

The explicit Premium production integration uses the calibrated lookup reference engine. It keeps the SAWSTAR name, plugin ID, parameter IDs, preset folder and state format. Preview and Candidate remain separate test identities. The default source build stays Classic until acceptance gates are met.

## Project migration

Classic state and preset data remain readable, but the Premium filter changes the sound. This is format compatibility, not identical Classic recall. Preserve the previous plugin and projects. Do not install Classic and this RC side by side with the same SAWSTAR identity.

## Filter and latency

LP12, LP24, HP12 and BP12 retain the existing controls. The Premium build reports 32 samples of latency; dry and wet filter paths are aligned. Check host compensation, automation and bypass with the exact RC. Calibration retains dry unity, the accepted wet gain and preset trims; it is not automatic loudness matching.

## Factory checks

The previous 24-preset RC headroom grid covered 44.1/48 kHz and 1/8/16 held roots at velocity 127. All 144 cells passed the unchanged -6 dBFS pre-guard limit, worst -6.51293 dBFS. The six added performance presets require the expanded 252-case grid: 180 default cases plus 72 full/swept wheel-pressure cases. See the exact build evidence; these are engineering checks, not musical listening or a realtime CPU guarantee.

## Open release gates

Robert approved listening to the six exported snapshots for the new presets; that does not accept this exact 1.2.1 RC. Native Mac/Windows CPU/dropout, filter quality, GUI/lifecycle and exact-RC host checks remain open, as does K1 numerical resolution. A validated ZIP is not native acceptance. No publication is authorized by this manual. The current public release is unchanged; this RC is not published.